

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458860.8561 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.186 +/- 0.055
Transit depth (Rp/Rs)^2	3.46 +/- 2.04 [%]
Orbital Inclination (inc)	78.88 +/- 2.89
Airmass coefficient 1 (a1)	0.9 +/- 0.54
Airmass coefficient 2 (a2)	0.18 +/- 0.37
Scatter in the residuals of the lightcurve fit is	68.4 %
Best Comparison Star	None
Optimal Aperture	2.1
Optimal Annulus	8.65
Transit Duration (day)	0.035 +/- 0.019

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.186 $\pm$ 0.055 vs 0.1594 (deviation 0.35 σ)
Duration fit vs published	0.035 d vs 0.0512 d (deviation 0.61 σ)
Mid-time O-C	-3.4 min
Depth SNR	2.4
Residual scatter	68.4 %

Light curve

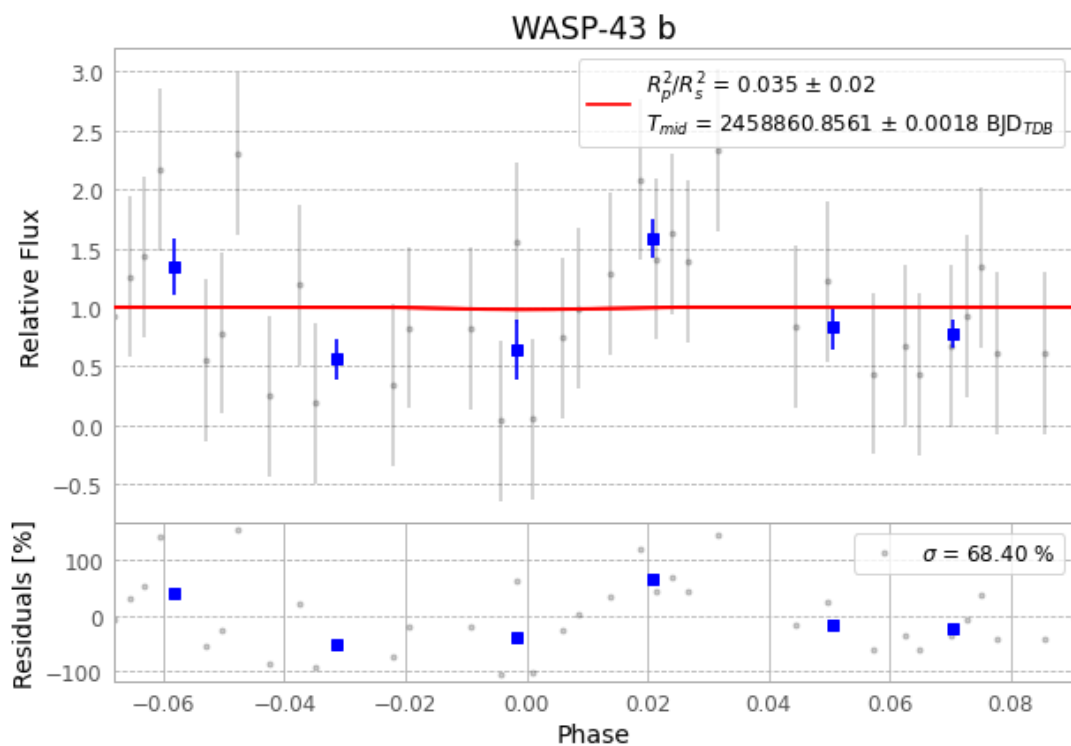


Figure 1 — FinalLightCurve_WASP-43 b_2020-01-12.png (SHA-256 cbb760388aaf9f2...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [242, 315], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b7a24e27a7d6c6f0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

64 FITS frames (42 MB), WASP-43200112070612.FITS → WASP-43200112101512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-200112.log (953876c782e8...), FinalLightCurve_WASP-43 b_2020-01-12.png (cbb760388aaf...), FinalLightCurve_WASP-43 b_2020-01-12.csv (33acad564866...)

Compute resources

Wall time 118.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-08 01:06Z.